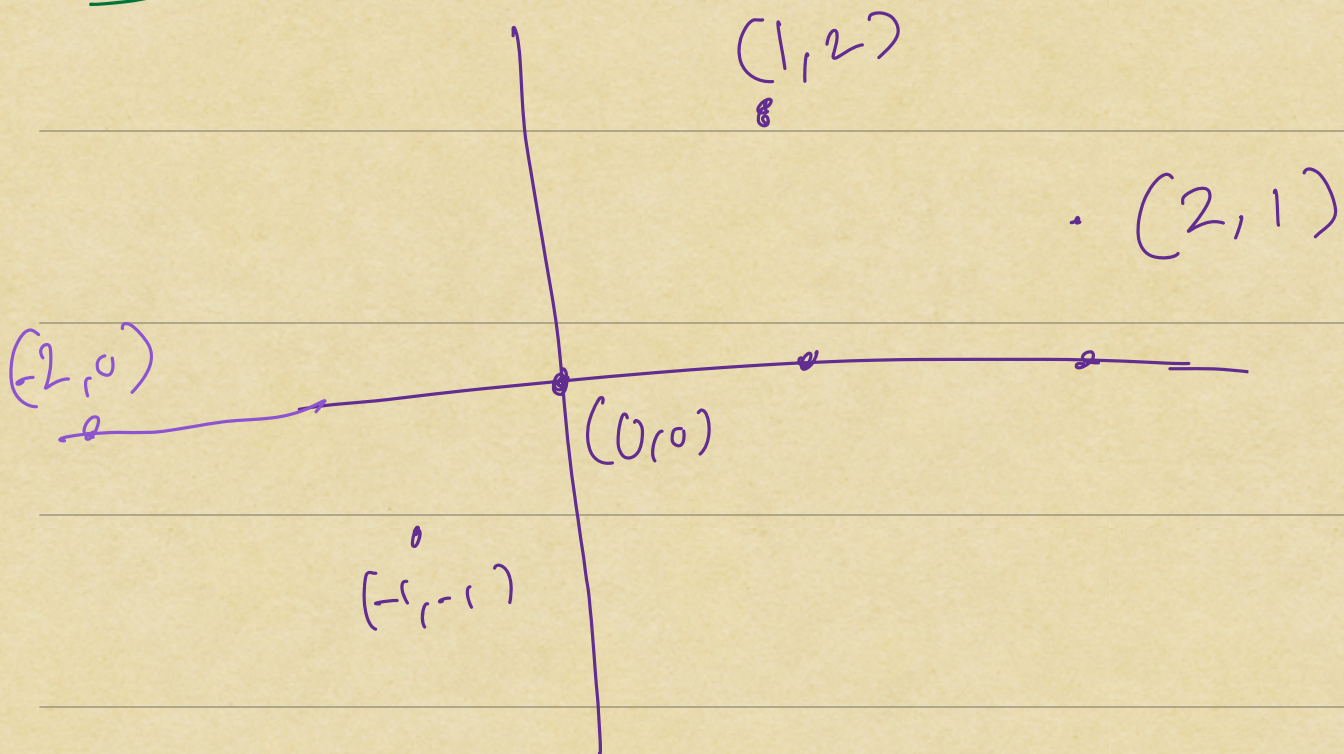


Orthogonality ($F = \mathbb{R}$)

Example:



Q: Is there a cubic fn whose graph hits these values?

A:

$$f(x) = a_3 x^3 + a_2 x^2 + a_1 x + a_0$$

Wanted: $f(-1) = -1$

$$f(0) = 0$$

$$f(1) = 2$$

$$f(2) = 1$$

$$f(-2) = 0$$

$$\Rightarrow -a_3 + a_2 - a_1 + a_0 = -1$$

$$a_0 = 0$$

$$a_3 + a_2 + a_1 + a_0 = 2$$

$$8a_3 + 4a_2 + 2a_1 + a_0 = 1$$

$$\Rightarrow \left[\begin{array}{cccc|c} -1 & 1 & -1 & 1 & -1 \\ 0 & 0 & 0 & 1 & 0 \\ 1 & 1 & 1 & 1 & 2 \\ 8 & 4 & 2 & 1 & 1 \end{array} \right]$$

$$\begin{bmatrix} -8 & 4 & -2 & -1 & 0 \end{bmatrix}$$

$$\begin{array}{l} R_1 \rightarrow -R_1 \\ R_2 \rightarrow R_1 \end{array} \left[\begin{array}{cccc|c} 1 & -1 & 1 & -1 & 1 \\ 0 & 0 & 0 & 1 & 0 \\ 0 & 2 & 0 & 2 & 1 \\ 8 & 4 & 2 & 1 & 1 \end{array} \right]$$

$$\begin{array}{l} R_3 - 8R_1 \\ R_2/2 \end{array} \left[\begin{array}{cccc|c} 1 & -1 & 1 & -1 & 1 \\ 0 & 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 1 & 1/2 \\ 0 & 12 & -6 & 9 & -7 \end{array} \right]$$

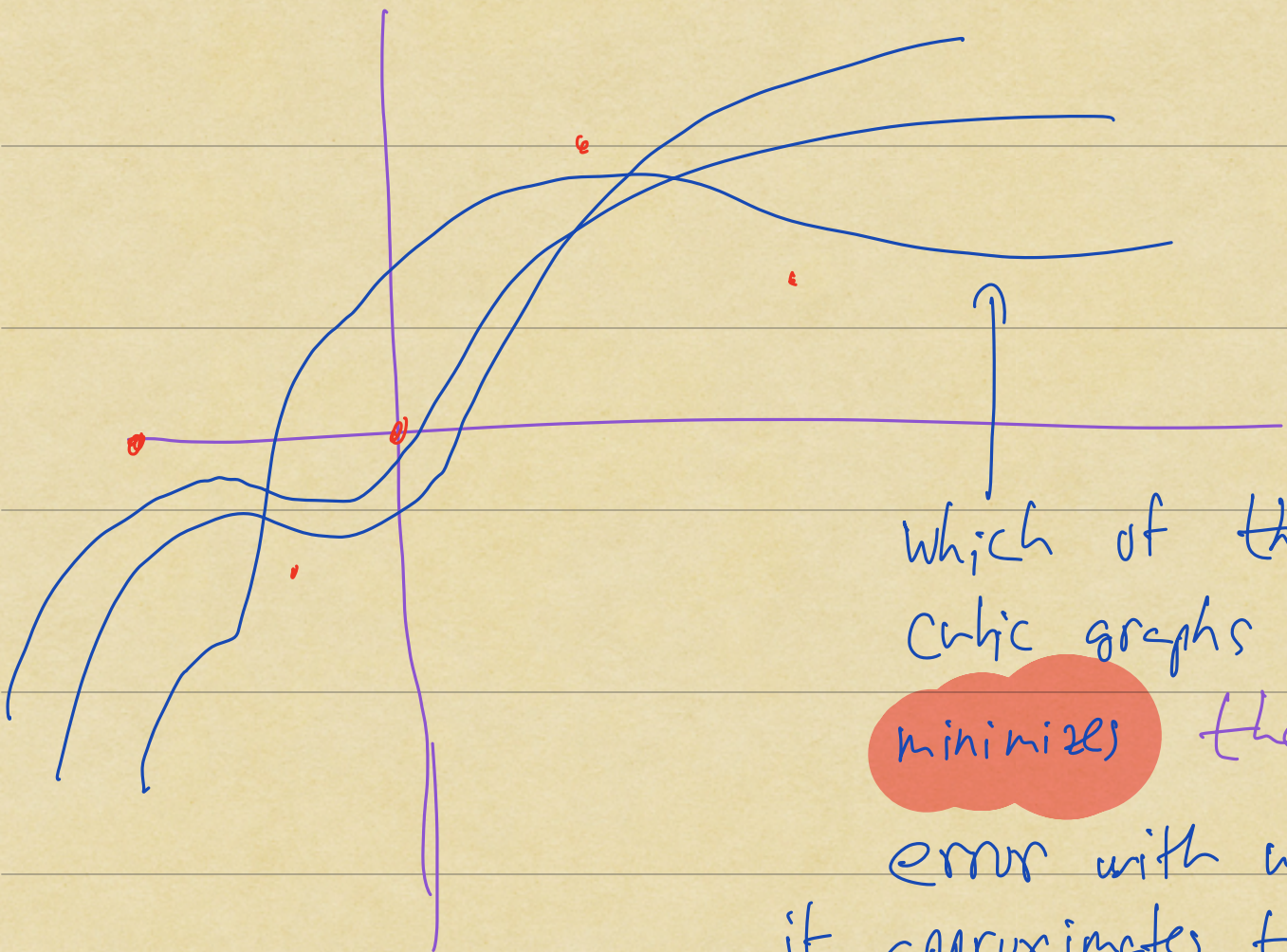
$$\underline{R_2 \leftrightarrow R_3} \left[\begin{array}{cccc|c} 1 & -1 & 1 & -1 & 1 \\ 0 & 1 & 0 & 1 & 1/2 \\ 0 & 0 & 0 & 1 & 0 \\ 0 & 12 & -6 & 9 & -7 \end{array} \right]$$

I can find a cubic fn fitting the

data; in fact there is a unique such fn.

If we add the additional condition $f(-2) = 0$, then

there is **no** cubic that fits the data **exactly**



Which of these cubic graphs **minimizes** the error with which it approximates the actual values?

Bird's eye view

$$Ax = v$$

(5×4 in example) (5×1 in example)

solve for x

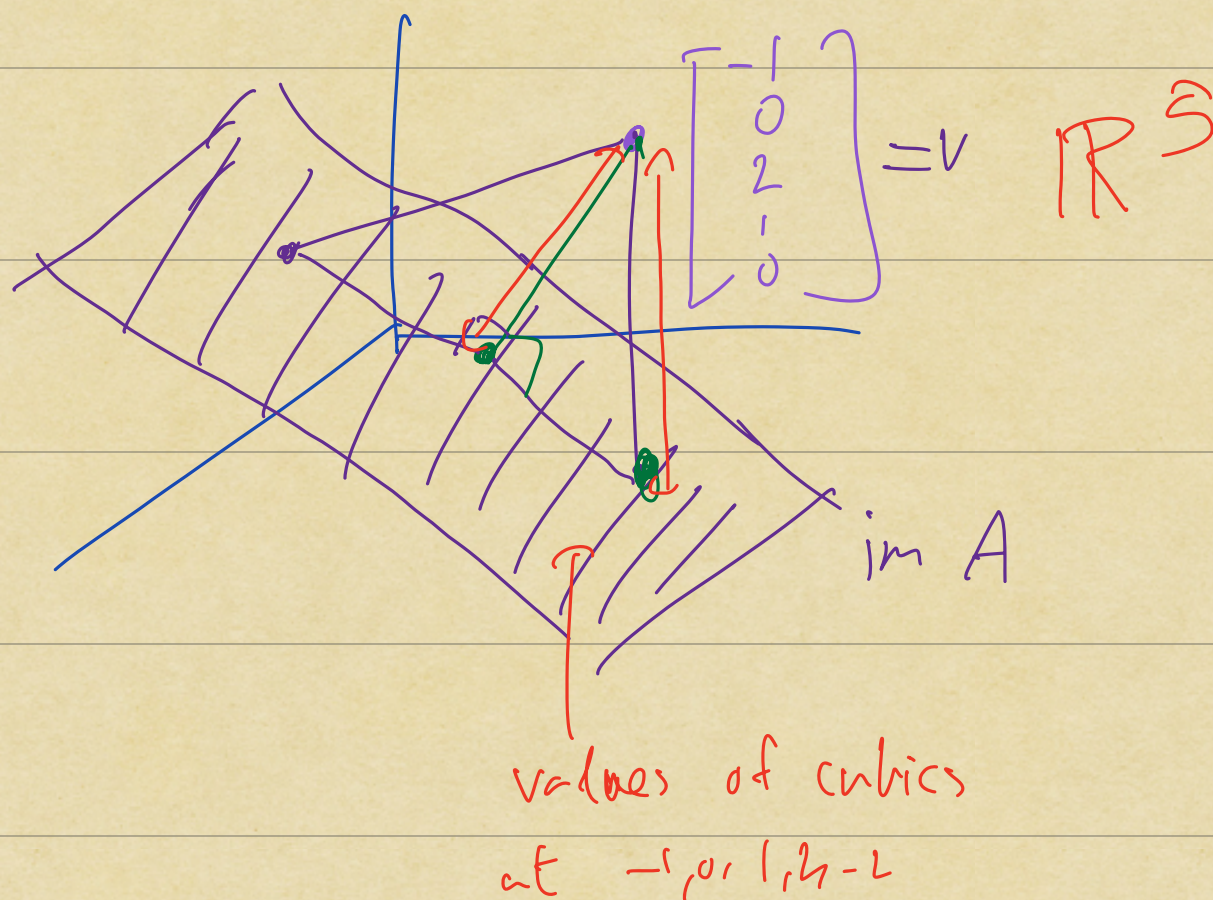
Saying this has no solutions

$$\Leftrightarrow v \notin \text{im } A$$

In the example, the image of 5×4 matrix A is the set of vectors in $\mathbb{R}^5$ that are of the form

$(f(-1), f(0), f(1), f(2), f(-2))$

for a cubic fn $f(x)$



The approximation will be given by the orthogonal projection of v onto $\text{im } A$.

Defⁿ (Dot product)

(1)

$$v, w \in F^n \leadsto v \cdot w = v^T w$$

$$\begin{bmatrix} a_1 \\ a_2 \\ \vdots \\ a_n \end{bmatrix} \cdot \begin{bmatrix} b_1 \\ b_2 \\ \vdots \\ b_n \end{bmatrix} = a_1 b_1 + a_2 b_2 + \dots + a_n b_n$$

(2) v & w are orthogonal

if $v \cdot w = 0$

Example:

$$(1) \quad e_i \cdot e_j = \begin{cases} 1 & \text{if } i=j \\ 0 & \text{otherwise} \end{cases}$$

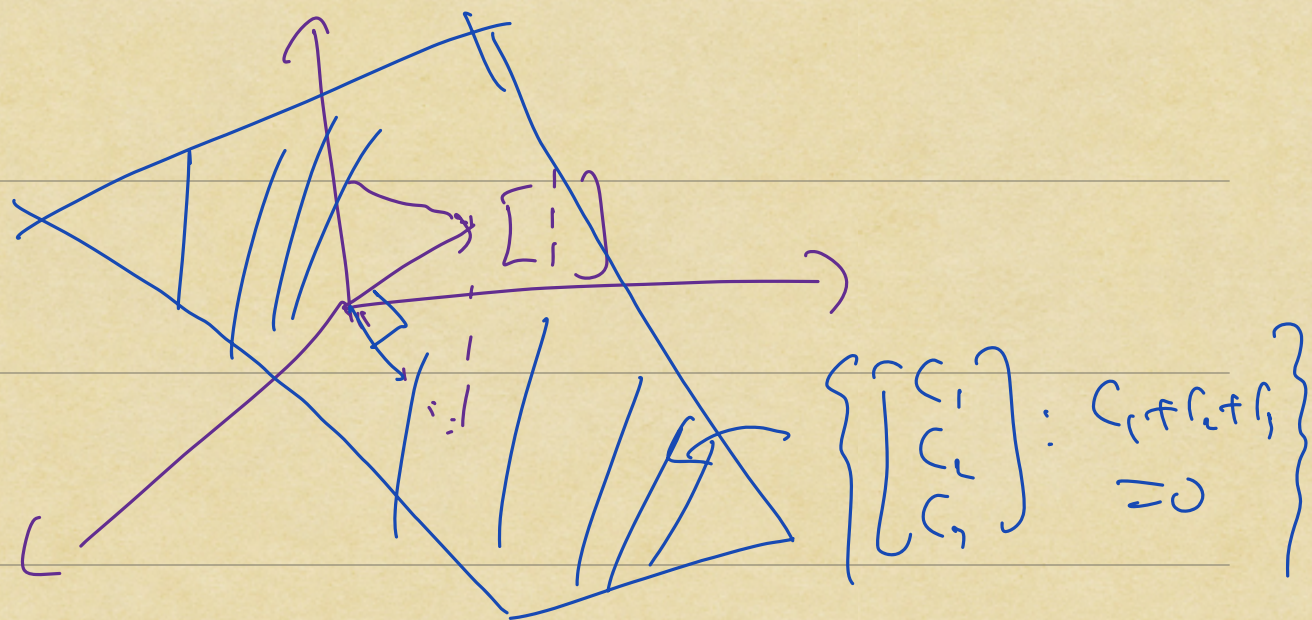
$$I_n \cdot I_n = I_n$$

Distinct standard basis vectors
are orthogonal to each other.

(2) $F = \mathbb{R}$

$$v = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}, w = \begin{bmatrix} c_1 \\ c_2 \\ c_3 \end{bmatrix}$$

$$v \cdot w = c_1 + c_2 + c_3$$



Observation: $W \leq F^n$ vector subspace

$$\left\{ v \in F^n : v \cdot w = 0, \forall w \in W \right\}$$

$\leq F^n$ is also a vector subspace.

Pf. This is because of two standard properties

$$\bullet v \cdot (cw) = c \cdot (v \cdot w)$$

$$v, w \in F^n, c \in F$$

$$\bullet v \cdot (w_1 + w_2) = v \cdot w_1 + v \cdot w_2$$